

Project Report: Qwen Fine-Tuning for CS/AI Paper Translation

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Abstract

This project aims to build a locally deployable English-to-Chinese translation model for computer science and artificial intelligence papers. The target behavior is accurate, fluent, formal academic Chinese with consistent terminology and robust handling of long technical sentences. The initial plan used full-parameter SFT of Qwen3-32B in multiple stages: general/scientific translation adaptation followed by CS/AI paper specialization. Two Qwen3-32B full-parameter runs have now been completed or partially completed: a Stage 1 run to the selected 1000-step checkpoint, and a smaller continuation run from that checkpoint to 1500 steps. The training evidence shows that validation loss reached its current best value at step 1000, while further training did not improve validation loss and manual inspection suggested that noisy public parallel corpora pushed the model toward a generic machine-translation style. The project therefore changes direction from multi-stage training on public corpora to a single-stage high-quality distillation setup: collect highly cited AI/CS arXiv papers, generate paper-style Chinese translations with a strong GPT-5.4 high teacher, and fine-tune Qwen3-4B once 50K high-quality examples are available.

1 Project Goal and Current Status

The target system is not a general translation demo. It is intended for paragraph-level translation of CS/AI academic papers. The model should:

- produce Simplified Chinese by default;
- use formal academic style rather than casual or literal machine-translation style;
- preserve equations, LaTeX commands, citations, URLs, code snippets, variables, and section numbers;
- translate AI/CS terminology consistently, with Chinese–English paired terms when useful;
- avoid explanations, summaries, refusals, self-talk, and `<think>` traces;
- remain deployable locally after fine-tuning.

The repository already contains a runnable engineering pipeline. Stage 1 data preparation, SFT chat conversion, Qwen3-32B full-parameter training, DeepSpeed ZeRO-3 configuration, W&B logging, arXiv/OpenAlex paper collection, HTML segment extraction, teacher translation, and a basic SacreBLEU evaluator are implemented. The environment is managed with `uv`. The main dependencies include `torch`, `transformers`, `trl`, `deepspeed`, `datasets`, `sacrebleu`, and `wandb`.

2 Implemented Pipeline

2.1 Repository Components

The current repository is organized around data processing, training, evaluation, and model assets:

- `zero3_bf16.json`: BF16 DeepSpeed ZeRO-3 runtime configuration.
- `qwen3_32b_stage1_full.yaml`: main Qwen3-32B Stage 1 training config.
- `qwen3_32b_stage1_bestval_from1000.yaml`: continuation config.
- `scripts/training/train_sft.py`: training entry point using Transformers and TRL `SFTTrainer`.

- `src/nlp_project/data/processing.py`: Stage 1 filtering, normalization, and SFT message construction.
- `src/nlp_project/data/stage2_collection.py`: OpenAlex/arXiv discovery and citation filtering.
- `src/nlp_project/data/stage2_segments.py`: arXiv HTML cleaning and segment extraction.
- `stage2_translation.py`: Responses-compatible teacher translation client and SFT record builder.
- `evaluate_translation.py`: current lightweight evaluation entry point.

The training entry point enables assistant-only loss, gradient checkpointing, BF16, packing, and an attention implementation fallback. The DeepSpeed config uses ZeRO-3 parameter sharding, overlapped communication, contiguous gradients, automatic batch parameters, and BF16. This confirms that the 32B full-parameter path is engineered around sharded training rather than naive DDP.

2.2 Data Format

The project uses a clean intermediate JSONL schema:

```
{
  "id": "...",
  "source": "English source text",
  "target": "Chinese translation",
  "domain": "general|scientific|cs_ai_paper",
  "split": "train|validation|test",
  "metadata": {...}
}
```

Each example is then converted into a three-message SFT record: a system translation instruction, a user translation request, and the assistant Chinese translation. The current filters remove empty or very short examples, obvious language mismatches, severe length-ratio outliers, Traditional Chinese or non-Chinese-script targets, self-talk, refusals, and thinking tags.

3 Stage 1 Data Analysis

Table 1 summarizes the data used for the completed Stage 1 experiments and the current arXiv segment pool. Stage 1 has a large number of examples, but 1M of the training samples come from QuickMT and 158.6K come from CSL abstracts. The Stage 1 training source length is short: the mean is 194.8 characters and the median is 82 characters. This is far from the target use case, where paper paragraphs often contain long clauses, citations, equations, and dense terminology. By contrast, the extracted Stage 2 arXiv segments average 943.0 source characters.

Table 1: Main dataset statistics used in this report.

Dataset	Examples	Source / domain	Papers	Avg. source chars	Avg. target chars
Stage 1 train	1158637	1M QuickMT + 158.6K CSL	158637	194.8	64.5
Stage 1 validation	10683	683 QuickMT + 10K CSL	10000	821.4	225.6
Stage 1 test	10667	667 QuickMT + 10K CSL	10000	835.6	229.7
Stage 2 segments	93555	arXiv CS/AI papers	1663	943.0	-

The main issue is data distribution, not data volume:

1. QuickMT is broad-domain and does not consistently match academic CS/AI style.
2. CSL is closer to scientific writing, but it is mostly title-and-abstract data rather than full paper paragraphs.
3. Public parallel corpora contain substantial machine-translation artifacts.
4. Qwen3-32B already has strong translation ability, so low-quality targets can pull the model toward a worse translation style instead of improving it.

A representative QuickMT example shows the mismatch. It is not necessarily wrong, but it is not useful enough for teaching CS/AI paper style:

Source: Of all these great limitations and frameworks which fashion and create the poetry and variety of life, the family is the most definite and important.

Target: 在塑造和创造着生活这首诗歌及其斑斓色彩的这些所有伟大的限制和框架中，家庭是最明确、最重要的限制和框架。

4 Qwen3-32B Full-Parameter Experiments

4.1 Training Setup

The first run fine-tuned `assets/models/Qwen3-32B` on Stage 1 SFT data. The main settings were: 8 GPUs, BF16, DeepSpeed ZeRO-3, max sequence length 4096, per-device batch size 2, gradient accumulation 4, learning rate 5×10^{-6} , cosine schedule, warmup ratio 0.03, weight decay 0.05, gradient clipping 1.0, gradient checkpointing enabled, and assistant-only loss enabled.

The second run resumed from `checkpoint-1000` under the Stage 1 checkpoint directory. It kept the same core optimization settings, but changed the total target to 1500 steps and used denser save/eval/logging intervals to test whether a small continuation from the selected checkpoint could improve validation performance.

Table 2: Completed and partially completed Qwen3-32B training runs.

Experiment	Source	Steps	Train loss	Eval loss	Eval token acc.	Status
Stage1 full	checkpoint-1000	1000	0.8365	0.6761	0.8225	best eval point
From1000 small LR	checkpoint-1500	1500	0.7443	0.6862	0.8215	no improvement

4.2 Training Evidence

Figure 1 shows validation loss over training steps. The Stage 1 run improved steadily through 1000 steps and reached the current best validation loss of 0.6761 at checkpoint-1000. The continuation run from step 1000 stayed around 0.6849–0.6876 between steps 1050 and 1500. The online W&B summary for the longer Stage 1 run also reports validation loss of 0.6990 at step 1480, which is worse than the selected 1000-step checkpoint.

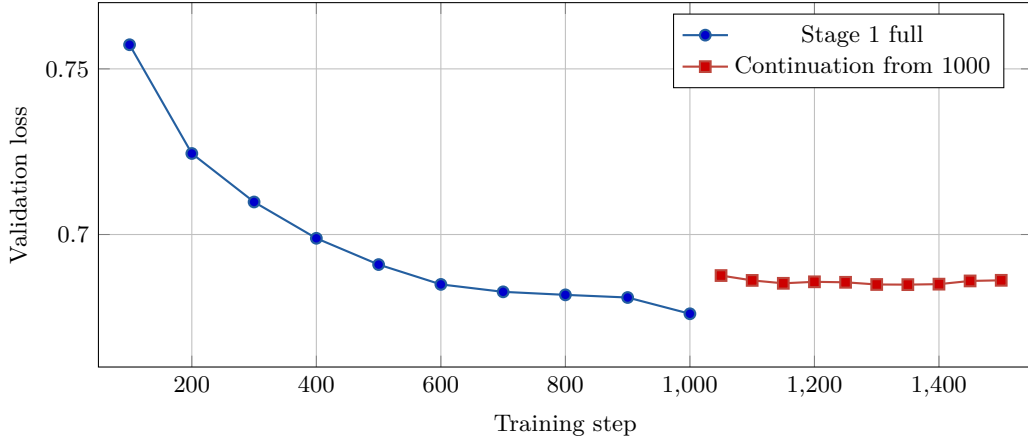


Figure 1: Validation loss for the Stage 1 run and the continuation run. The best recorded point is checkpoint-1000.

Figure 2 shows training loss. The continuation run can still reduce training loss, ending around 0.7443, but this does not translate into better validation loss. This separation suggests that additional training mainly fits the Stage 1 public-data distribution rather than improving the target paper-translation behavior.

4.3 Experiment Conclusions

The 32B training experiments support three conclusions.

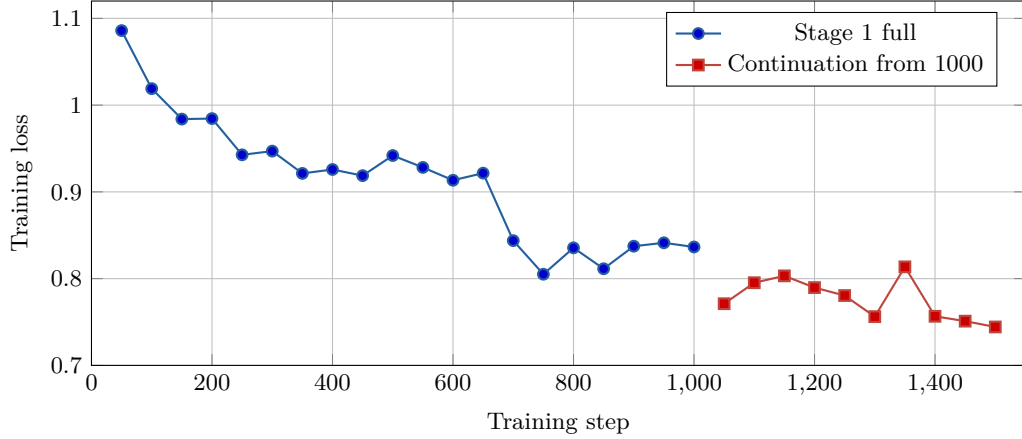


Figure 2: Training loss. More optimization on Stage 1 data lowers train loss but does not recover the best validation loss.

First, the full-parameter pipeline works. The repository has successfully run Qwen3-32B with BF16, DeepSpeed ZeRO-3, gradient checkpointing, TRL SFT, W&B logging, and saved checkpoints.

Second, continued Stage 1 training has low marginal value. Checkpoint-1000 is the best validation point currently available. Neither the longer Stage 1 run nor the smaller continuation run improves on it.

Third, data quality is now the main bottleneck. Validation loss measures fit to the Stage 1 validation distribution, not paper-translation quality. Manual inspection indicates that noisy public targets can make the model sound more like a generic machine translation system. Since Qwen3-32B already translates well, low-quality SFT targets can cause regression in style and fluency.

5 Change of Direction

The project is therefore moving from a multi-stage public-corpus recipe to a single-stage high-quality distillation recipe. Figure 3 summarizes the change.

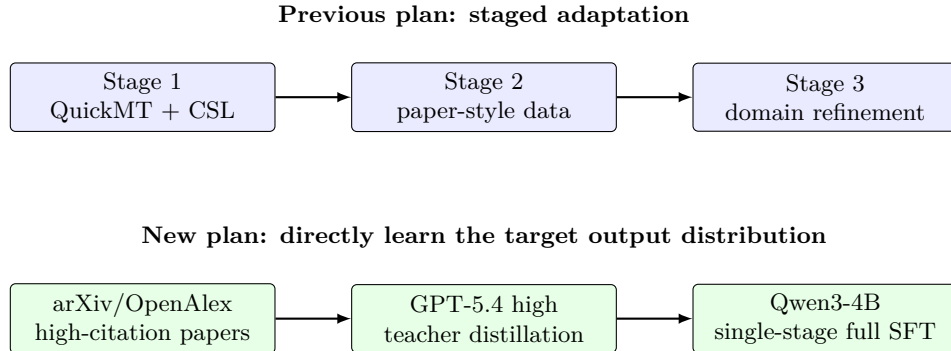


Figure 3: Training strategy change. The new plan reduces exposure to noisy public translations and focuses on high-quality paper-style targets.

The rationale is practical:

- The model does not need basic translation ability as much as it needs stable paper-translation style.
- Qwen3-32B is strong enough that low-quality SFT data can degrade its output distribution.
- Qwen3-4B is more realistic for local deployment and faster iteration.
- A strong teacher can define the desired translation style more directly than mixed public corpora.
- A single-stage setup simplifies ablation, data auditing, and evaluation.

6 New Data Progress

The Stage 2 collection pipeline has already produced a large candidate pool. The local OpenAlex/arXiv accepted-paper file contains 30,088 papers with citation count at least 21. The median citation count is 40, the mean is 85.3, and the maximum is 21,641. The HTML segment extractor has produced 93,555 English paper segments covering 1,663 papers. According to the current project status, about 40,000 teacher-distilled examples have been collected, and full-parameter Qwen3-4B training will start after reaching 50,000 examples.

Table 3: Stage 2 collection and extraction statistics.

Metric	Value
Accepted OpenAlex/arXiv papers	30088
Minimum citation count	21
Median citation count	40.0
Mean citation count	85.3
Maximum citation count	21641
Extracted English segments	93555
Papers covered by segments	1663

Figure 4 shows the accepted-paper year distribution. The pool is concentrated in 2020–2022.

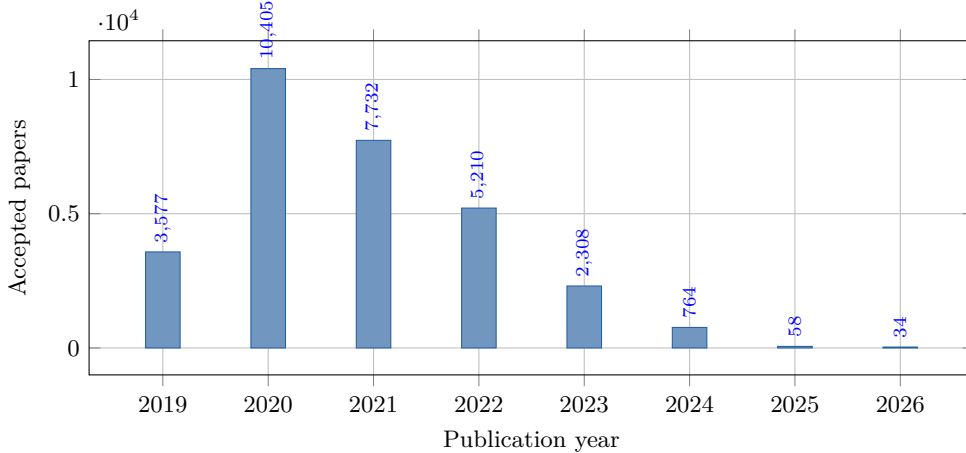


Figure 4: Year distribution of accepted OpenAlex/arXiv papers.

Figure 5 shows the top OpenAlex topics. The pool includes relevant AI topics such as NLP, multimodal learning, domain adaptation, and adversarial robustness, and it also contains many quantum-computing and quantum-information papers.

An extracted paper segment is much closer to the target task than the Stage 1 public examples:

Source: We introduce the concept of weighted rules under the stable model semantics following the log-linear models of Markov Logic. This provides versatile methods to overcome the deterministic nature of the stable model semantics, such as resolving inconsistencies in answer set programs, ranking stable models, associating probability to stable models, and applying statistical inference to computing weighted stable models.

A teacher translation sample:

我们引入了稳定模型语义 (stable model semantics) 下的加权规则 (weighted rules) 概念, 该概念遵循马尔可夫逻辑 (Markov Logic) 的对数线性模型 (log-linear models)。这提供了多种灵活的方法来克服稳定模型语义的确定性特征, 例如解决答案集程序 (answer set programs) 中的不一致性、对稳定模型进行排序、为稳定模型赋予概率, 以及将统计推断 (statistical inference) 应用于加权稳定模型的计算。

This is the desired output pattern: natural Chinese prose, English term retention where helpful, no extra explanation, and no hidden reasoning.

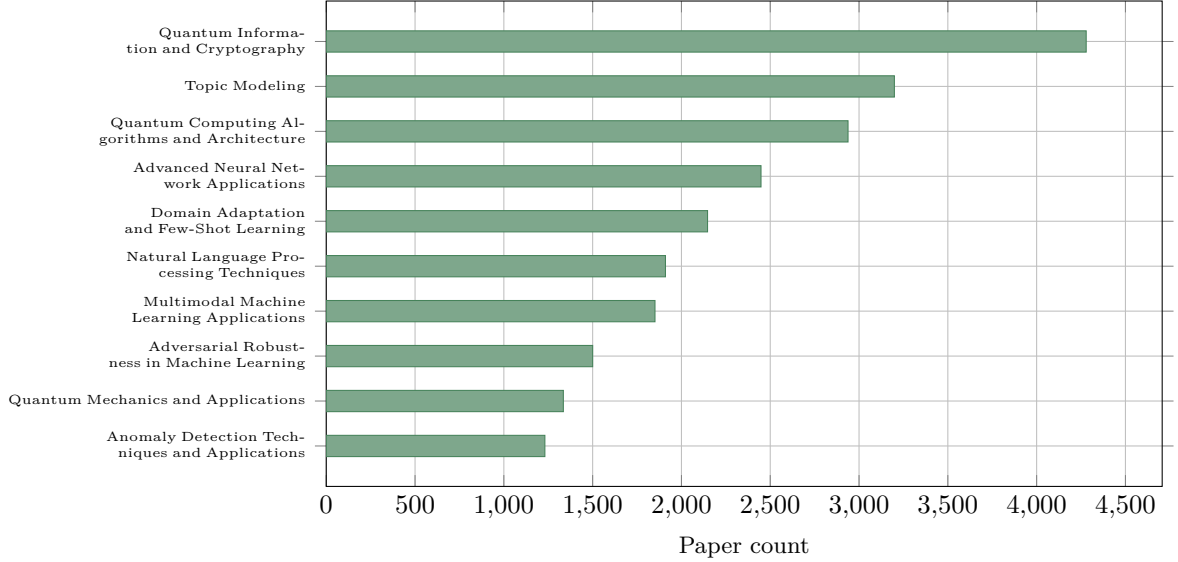


Figure 5: Top OpenAlex topics in the accepted paper pool. Some off-target topics should be downweighted.

7 Next Plan

7.1 Data Finalization

The next data milestone is to close the 50K distilled-example target. Before training, the data should be audited for:

- empty translations, self-talk, refusals, and thinking tags;
- unusually high English leakage in the Chinese target;
- severe source-target length-ratio outliers;
- malformed LaTeX, broken citations, or corrupted code snippets;
- paper-level train/validation/test leakage;
- off-target topics such as quantum physics or pure mathematics if they dominate the sample.

The split should be performed by paper id rather than by segment id. This prevents adjacent paragraphs from the same paper from appearing in both training and evaluation.

7.2 Qwen3-4B Single-Stage SFT

The proposed base model is `assets/models/Qwen3-4B-Instruct-2507`. The initial training recipe should use full-parameter SFT with:

- max sequence length 4096, with 8192 considered only after memory is stable;
- learning rate 5×10^{-6} to 1×10^{-5} ;
- cosine schedule and warmup ratio 0.03;
- per-device batch size 1 or 2 as the first safe setting;
- gradient checkpointing and BF16;
- assistant-only loss;
- evaluation every 50–100 steps during early experiments;
- a 20–50 step preflight run before any long training job.

7.3 Evaluation

The next evaluation suite should not rely only on Stage 1 validation loss. It should include:

1. SacreBLEU for compatibility with the current evaluator;
2. COMET or another semantic metric for meaning preservation;
3. terminology accuracy and consistency against an AI/CS glossary;
4. a small human evaluation set scored for adequacy, fluency, academic style, terminology, and format preservation.

Useful baselines are base Qwen3-4B, the single-stage Qwen3-4B SFT checkpoint, the saved Qwen3-32B on a small diagnostic subset, and teacher outputs.

8 Risks

- **Teacher style bias.** A strong teacher may impose a narrow translation style. Mitigation: diversify paper topics and segment types, and manually review samples.
- **Topic drift.** Citation thresholds overselect older or off-target papers. Mitigation: add topic and year sampling weights.
- **Data leakage.** Multiple segments from one paper can leak across splits. Mitigation: split by paper id.
- **Overfitting to annotation conventions.** The student may overuse English parentheses or fixed terminology patterns. Mitigation: evaluate terminology and fluency separately.

9 Conclusion

The result show that the engineering pipeline is usable, and Qwen3-32B full-parameter training has been demonstrated. However, the Stage 1 public corpora do not define the desired target distribution well enough. The selected 1000-step checkpoint is the best validation point, and further Stage 1 training does not justify its cost. The project should now focus on high-quality paper-style distillation and Qwen3-4B single-stage full-parameter SFT. The immediate priorities are to finish the 50K distilled dataset, audit topic and translation quality, run a short 4B preflight, and build a stronger evaluation suite centered on terminology, academic style, and human judgment.